2024

Georgia-Pacific Optimizes Operator Efficiency Using Generative AI on AWS

Learn how Georgia-Pacific, a leading manufacturer, is transforming knowledge management using Amazon Bedrock.

Overview | Opportunity | Solution | Outcome | AWS Services Used

Accelerated

operator onboarding

Captu

the knowledge of experienced operators

machine downtime

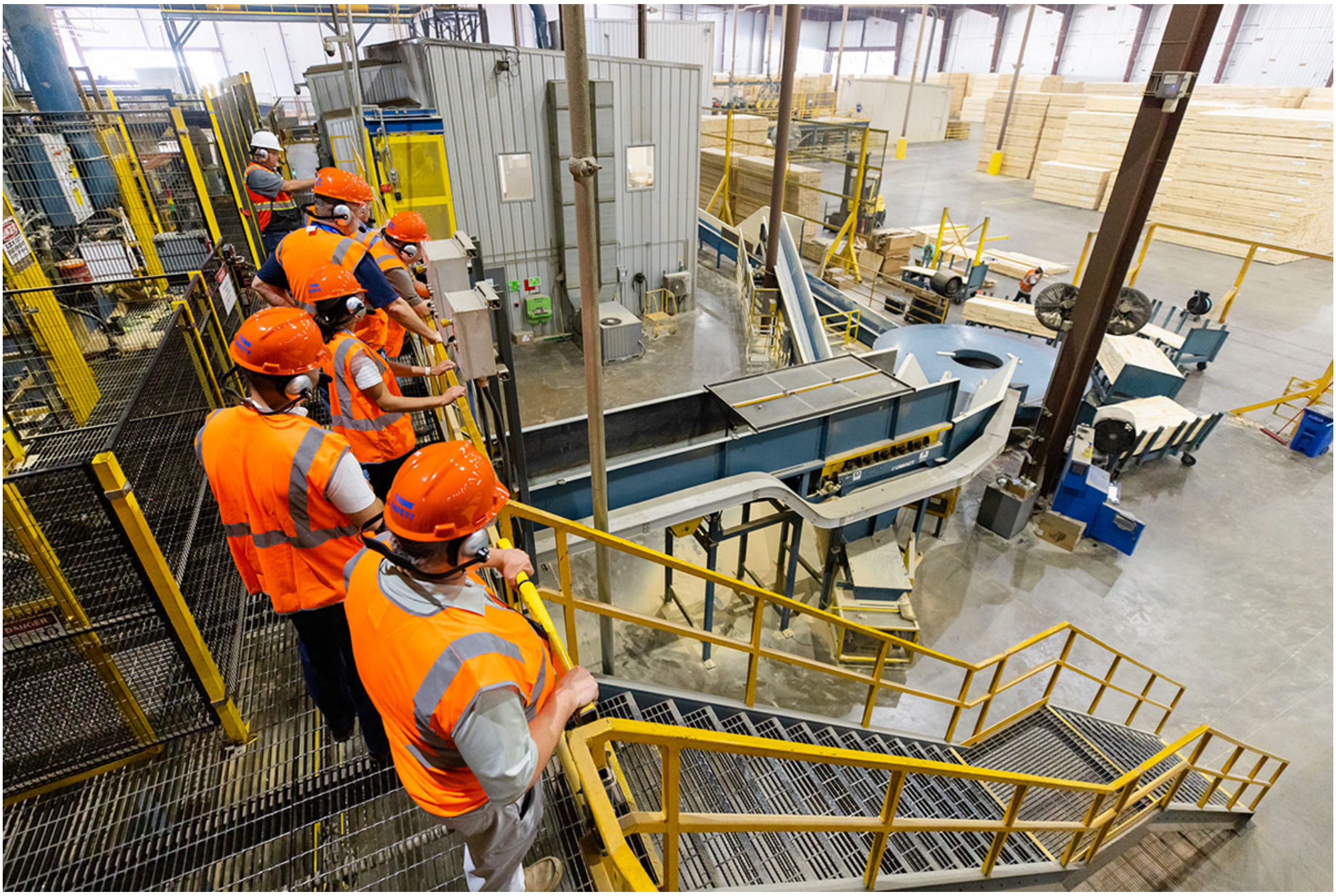
Reduced off quality

through enhanced troubleshooting

Overview

For nearly a century, Georgia-Pacific has been a major player in the manufacturing of paper products, packaging, and lumber. As a large-scale manufacturing operation, the company remains competitive by performing routine maintenance, optimizing manufacturing lines, and minimizing equipment downtime. However, the knowledge needed to properly carry out these critical functions was scattered across physical documents, digital files, and the minds of experienced employees.

To create a central source of information, Georgia-Pacific turned to Amazon Web Services (AWS) and the power of generative artificial intelligence (AI) to create a chatbot to respond to operators' questions. Georgia Pacific calls this chatbot ChatGP internally. With this vast, accessible repository of company knowledge, all Georgia-Pacific employees can get quick answers to complex queries that are accurately tailored to their processes.



Opportunity | Using Amazon Bedrock to Support Machine Operators for Georgia-Pacific

Founded in 1927, Georgia-Pacific is one of the largest manufacturers and distributors of pulp and paper products worldwide. The manufacturing company has over 140 facilities with a diverse range of equipment and processes. To keep operations running smoothly, junior operators and maintenance technicians need access to specialized knowledge of this equipment to optimize production and resolve issues quickly.

The lack of a centralized, readily accessible knowledge base often leads to reduced machine productivity, prolonged machine downtime, and higher troubleshooting and repair costs. As experienced employees who have spent decades running the production line retired or moved on, Georgia-Pacific risked losing valuab e company looked for a solution that could consolidate this disparate information, make it avail operators, and help preserve the expertise of the most knowledgeable employees for future g

“The longstanding issue of knowledge loss affects the entire manufacturing sector,” says Carter Smith, generative AI product leader at Georgia-Pacific. “Generative AI is helping us take proactive steps to solve it. As senior staff retire, implementing a reliable tool that has perfect memory empowers personnel and helps us gain and maintain our competitive advantage.”

A traditional approach to knowledge management would not suffice in a fast-paced, data-driven manufacturing environment. Making phone calls to experts and searching through binders, manuals, and maintenance records takes time and increases the risk of downtime. Georgia-Pacific chose to harness generative AI using Amazon Bedrock, a fully managed service that offers a choice of high-performing foundation models from leading AI companies.

"Our collaboration with the AWS team has equipped our operations with a straightforward and effective solution to enhance productivity and capture value ,” says Roshan Shah, vice president of applied AI and products at Georgia-Pacific. “In early 2023, when the world was learning about large language models, we took advantage of our data being hosted on AWS, and we chose to access the capabilities that we needed through Amazon Bedrock."

Solution | Transforming Knowledge Management with Generative AI

Georgia-Pacific engaged with AWS Professional Services, a global team of experts that helps companies realize their desired business outcomes on AWS. The teams collaborated to design and deploy a solution that would best meet the needs of its machine operators and maintenance technicians. The effort resulted in ChatGP, a chatbot powered by generative AI technologies.

Through Amazon Bedrock, Georgia-Pacific adopted Anthropic’s Claude, a large language model, to help the chatbot understand and respond to user queries with high accuracy and context. The company also uses Amazon Kinesis, a fully managed service that cost-effectively processes and analyzes streaming data at scale. By combining generative AI with Internet of Things data, ChatGP can provide more accurate and contextual responses to machine operators’ queries. For example, when an operator asks about a specific machine issue, ChatGP can reference the relevant Internet of Things data to provide insights into the machine’s current state, recent performance trends, and potential causes of the problem, thereby combining text and numbers in one response.

Machine operators interact with ChatGP through a user-friendly web application accessible from any computer or tablet on the Georgia-Pacific network. The chatbot serves as a centralized knowledge hub, consolidating information from various sources, including digital documents, maintenance records, and Inte nsor data. When operators encounter an issue, they can ask ChatGP for assistance and receive step e on machine inputs or adjustments. ChatGP adapts to the needs of each facility so that the information is tailored to the equipment and processes used at that location.

Georgia-Pacific prioritized capturing the knowledge of experienced workers. “We have equipment that’s 50 years old, and for many of these machines, we lack proper documentation about operating procedures. That knowledge exists in the minds of our experienced employees,” says Shah. “We’ve built a capability where we can record a conversation between multiple subject matter experts or retired employees, and a large language model summarizes the discussion. The output is a document that looks as if it were created by the manufacturer, and the relevant knowledge becomes instantly available to all applications referring to the large language model.”

The implementation of ChatGP yielded significant benefits for Georgia-Pacific. The company has improved machine production by giving operators the data that they need to make near real-time adjustments. This has reduced off quality, minimized machine downtime, and boosted productivity. “Using generative AI to pull together information from various sources for quick and easy answers has been a game-changer for us,” says Ryan Holbird, senior manager of digital transformation at Georgia-Pacific. “It’s like having a digital expert who’s always there, ready to help both new and seasoned operators find the right answers fast.”

Outcome | Improving Knowledge Transfer and Efficiency in Large-Scale Manufacturing

By harnessing the power of generative AI, Georgia-Pacific set a new standard for knowledge management and issue resolution in large-scale manufacturing. “We’re just beginning, but we estimate millions in potential savings annually across our facilities through the use of generative AI technology,” says Shah.

Looking ahead, Georgia-Pacific plans to expand ChatGP to even more of its facilities by the end of 2024.

“It was inspiring to witness our teams coming together to solve this problem,” says Shah. “There has been a remarkable business impact in a short time, primarily driven by the talented teams at Georgia-Pacific, Koch, and AWS. The products created by this team stem from their self-actualization and commitment to improving the lives of our operators. I look forward to what the future holds for this technology at our company.”

About Georgia-Pacific

Founded in 1927 and based in Atlanta, Georgia, Georgia-Pacific is one of the largest manufacturers and distributors of pulp and paper products worldwide. Its brands include Angel Soft, Dixie Insulair, Vanity Fair, Brawny, Sparkle, and more.

AWS Services Used

Amazon Bedrock

The easiest way to build and scale generative AI applications with foundation models.

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AWS Professional Services

The AWS Professional Services organization is a global team of experts that can help you realize your desired business outcomes when using the AWS Cloud.

Learn more »

Amazon Kinesis

Amazon Kinesis cost-effectively processes and analyzes streaming data at any scale as a fully managed service.

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JAPAN

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2024

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KABAM Robotics created Smart+ – a cloud-based robot management platform which leverages AWS to enhance the intelligence and concierge capabilities of their robots – and REMI – a groundbreaking system equipping their robots with the power of large language models, seamlessly integrated using Amazon Bedrock API, while

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